

Project Title

Multi-Modal University Chatbot System

Description

Developed a smart university chatbot system in Python that supports **text, voice, and hybrid interaction modes**. The chatbot uses **speech recognition, text-to-speech conversion, spam detection, and intent recognition** to assist users with university-related queries such as admissions, fees, courses, and schedules.

Technologies Used

- Python
- SpeechRecognition
- Pyttsx3
- Google Speech API
- Natural Language Processing (Keyword-Based Intent Detection)

Key Features

- Multi-modal interaction (Text + Voice)
- Speech-to-Text using microphone input
- Text-to-Speech voice responses
- Spam query detection
- Intent recognition with confidence scoring
- Dynamic and static response generation
- Hybrid mode for combined interaction
- Menu-driven chatbot system

Functional Modules

- Spam Detection Module
- Intent Identification Module
- Dynamic Response Generator
- Voice Interaction Module
- Text Interaction Module
- Hybrid Interaction Module

Supported Intents

- Admission Queries
- Fee Information

- Courses & Programs
- Schedule & Exam Information

Sample Functionalities

- Detects spam-related queries automatically
- Answers admission eligibility questions
- Provides fee structure information
- Explains university programs and courses
- Responds through voice output
- Accepts voice commands using microphone input

Learning Outcomes

- Human Computer Interaction (HCI)
- Voice-based User Interfaces
- Natural Language Processing Basics
- Python-based AI Applications
- Multi-modal System Design
- Speech Recognition Integration

Role

Designed and implemented the complete chatbot system including:

- Voice interaction handling
- Intent classification
- Spam detection logic
- Dynamic response generation
- User interaction flow design

Conclusion

The Multi-Modal University Chatbot System successfully demonstrates the integration of text and voice-based interaction within a single intelligent application. The project provides an efficient way for users to access university-related information through natural communication methods. By combining speech recognition, text-to-speech conversion, spam detection, and intent recognition, the chatbot improves user experience and accessibility.

This project enhanced practical understanding of Human Computer Interaction (HCI), Natural Language Processing concepts, and Python-based application development. It also highlighted the importance of designing interactive systems that are user-friendly, responsive, and capable of handling multiple communication modes effectively.